

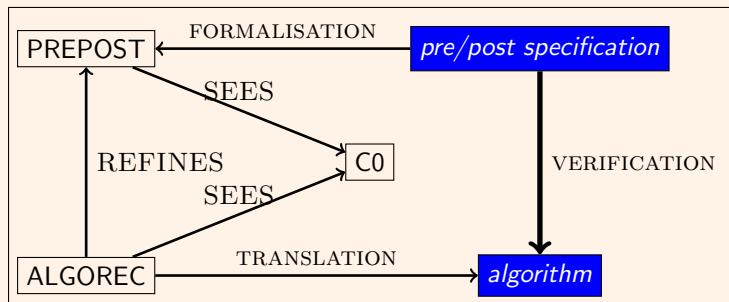
- 1 CM5
- 2 The Call-as-Event Paradigm

① CM5

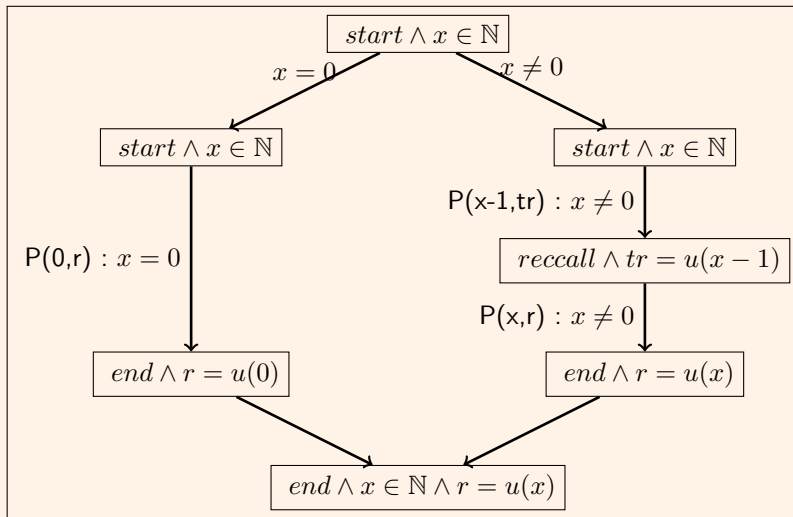
② The Call-as-Event Paradigm

- ## ② The Call-as-Event Paradigm

The recursive pattern



The refinement diagram



Ideas for producing algorithms

- **EVENT** $\text{rec}\% \text{PROC}(h(x), y) \% P(y)$ is simply simulating the recursive call of the same function.
- The invariant is defined in a simpler way by analysing the inductive structure and a control variable is introduced for structuring the inductive computation.

```
EVENT
e
WHERE
   $\ell = \ell_1$ 
   $g_{\ell_1, \ell_2}(x)$ 
THEN
   $\ell := \ell_2$ 
   $x := f_{\ell_1, \ell_2}(x)$ 
END
```

```
EVENT
   $\text{rec}\% \text{PROC}(h(x), y) \% P(y)$ 
ANY y
WHERE
   $\ell = \ell_1$ 
   $g_{\ell_1, \ell_2}(x, y)$ 
THEN
   $\ell := \ell_2$ 
   $x := f_{\ell_1, \ell_2}(x, y)$ 
END
```

```
EVENT
   $\text{call}\% \text{APROC}(h(x), y) \% P(y)$ 
ANY y
WHERE
   $\ell = \ell_1$ 
   $g_{\ell_1, \ell_2}(x, y)$ 
THEN
   $\ell := \ell_2$ 
   $x := f_{\ell_1, \ell_2}(x, y)$ 
END
```

The computation of the function x^2

VARIABLES

r, c, tr

INVARIANTS

$inv1 : r \in \mathbb{N}$

$inv2 : c = end \Rightarrow r = n * n$

$inv3 : c = callrec \Rightarrow n \neq 0$

$inv4 : c = callrec \Rightarrow tr = (n - 1) * (n - 1)$

$inv5 : c \in C$

$inv6 : tr \in \mathbb{N}$

$inv7 : c = end \Rightarrow r = n * n$

$inv8 : c = end \wedge n \neq 0$

$\Rightarrow tr = (n - 1) * (n - 1) \wedge r = tr + 2 * (n - 1) + 1$

$inv9 : c = callrec \Rightarrow n * n = tr + 2 * (n - 1) + 1$

The computation of the function x^2

```
EVENT INITIALISATION
BEGIN
  act1 :  $r := 0$ 
  act2 :  $c := start$ 
  act3 :  $tr \in \mathbb{N}$ 
END
EVENT square0
REFINES square( $n$ ;  $r$ )
WHEN
  grd1 :  $c = start$ 
  grd2 :  $n = 0$ 
THEN
  act1 :  $c := end$ 
  act2 :  $r := 0$ 
END
```

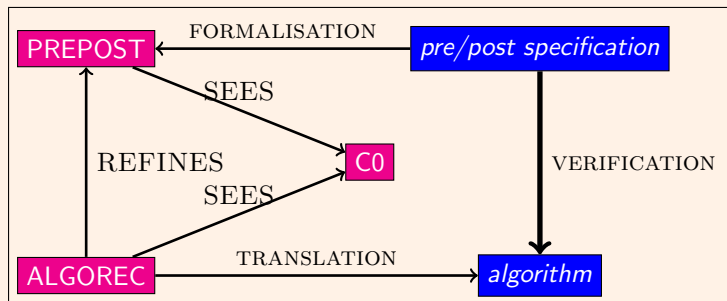
```
EVENT squaren
REFINES square( $n$ ;  $r$ )
WHEN
  grd1 :  $c = callrec$ 
THEN
  act1 :  $r := tr + 2 * (n - 1) + 1$ 
  act2 :  $c := end$ 
END
EVENT rec%square( $n-1$ ;  $tr$ )
WHEN
  grd1 :  $c = start$ 
  grd2 :  $n \neq 0$ 
THEN
  act1 :  $c := callrec$ 
  act2 :  $tr := (n - 1) * (n - 1)$ 
END
```

Summary for proof obligations

Summary for proof obligations

Name	Total	Automatic	Interactive
cae-square	34	32	2
square0	3	2	1
specsquare	2	2	0
square	29	28	1

The recursive pattern



- Proofs are easier and simpler
- Invariant is simple to find
- Translation is automatic
- Students are not happy with it and tools are not always set for this verification.